

Mahriar Gharaghani

Research CV — Artificial Intelligence / Decision Intelligence

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Research Profile

Research-oriented AI/ML engineer with interests at the intersection of artificial intelligence, sequential decision-making, and resource allocation. Current graduate research focuses on AI-driven project portfolio budgeting under uncertainty and dynamic cash-flow constraints, combining simulation-based evaluation, optimization benchmarks, and reinforcement learning. Prior experience in project control and project management environments provides practical grounding for research questions in planning, feasibility, and decision support.

Research Interests

- Reinforcement learning and sequential decision-making under uncertainty
- Optimization-informed AI and simulation-based benchmarking
- Decision-support systems for project portfolio management and constrained budgeting
- Retrieval-augmented generation (RAG) and LLM-based workflow tools

Education

M.Sc. in Engineering Management, Iran University of Science and Technology (IUST) 2023–2026

Tehran, Iran — In Progress

Thesis: *Autonomous Budgeting AI Agent*. Research on project portfolio budget allocation under uncertainty and dynamic financial constraints using a Python-based deep reinforcement learning framework. The work is evaluated against a perfect-foresight optimization benchmark as an upper-bound reference. The thesis context has received external support under contract with **Tehran Industrial Estates Company**.

B.Sc. in Industrial Engineering, Isfahan University of Technology (IUT) 2017–2023

Isfahan, Iran

Undergraduate education in industrial engineering with emphasis on quantitative analysis, planning, and systems thinking.

Ph.D. Entrance Exam, Computer Engineering (AI & Robotics) 2026

National Universities of Iran — Results Pending

Preparation included implementation-oriented study in data structures and algorithms, machine learning, and pattern recognition. Developed **RAG4Konkur**, a local AI study system for flashcard generation, concept review, and AnkiDroid-ready exports.

Research Experience

M.Sc. Thesis Researcher 2023–2026

Iran University of Science and Technology

Formulating project portfolio budgeting as a sequential decision problem under uncertainty and

dynamic cash-flow constraints. Developing a Python-based experimental framework including environment design, policy learning, and evaluation procedures. Comparing learned policies against a perfect-foresight optimization solution as an upper-bound benchmark.

Industry-Supported Research Context

2023–2026

Tehran Industrial Estates Company

Research problem received external institutional support, reflecting the practical relevance of AI-assisted budgeting and planning in real project portfolio environments.

Selected Research and Technical Projects

Autonomous Budgeting AI Agent

2023–Present

Research / Thesis System

Developed a research-oriented prototype integrating simulation, optimization baselines, and reinforcement learning for feasible budget allocation under uncertainty.

RAG4Konkur

2025–2026

Local Retrieval-Augmented Study System

Built a local RAG pipeline for concept retrieval, flashcard generation, review workflows, and AnkiDroid-ready export.

House Price ML Pipeline

2024

Machine Learning Workflow

Implemented preprocessing, feature engineering, model training, evaluation, and structured reporting in an end-to-end supervised learning pipeline.

PyTorch CIFAR-10 Pipeline

2024

Deep Learning Experimentation

Implemented an image classification workflow using PyTorch, including data loading, augmentation, checkpointing, and ResNet18 transfer learning.

Model Serving with FastAPI and Docker

2024

Deployment / ML Systems

Exposed a machine learning model through a FastAPI-based REST API and containerized the service using Docker.

Robotic Arm RL Simulation

2024

Exploratory Reinforcement Learning

Explored reinforcement learning behavior in a simulated control task to strengthen understanding of sequential control and policy learning.

Decision-Support and Visualization Prototypes

2023–Present

Applied Technical Projects

Developed practical prototypes including logistics heatmaps, live market visualizations, spreadsheet-style utilities, and workflow-oriented AI tools.

Professional Experience

Project Management Officer, Iran Energy Industries Corporation (IEIC)

2023

Tehran, Iran

Worked in a project-based organizational environment with responsibilities related to coordi-

nation, planning support, and structured reporting. Served as senior deputy to the Head of Department.

Project Control Specialist, Jahanpars Group

2019–2020

Iran

Contributed to project control activities in a large industrial setting while still an undergraduate student. This role shaped later research interest in planning, uncertainty, and resource allocation.

Independent Technical Reorientation

2023–Present

Self-Directed Transition

Undertook a focused transition toward technical and research-driven AI systems development, with sustained work in software engineering, machine learning pipelines, deployment practices, and AI applications.

Technical Skills

Programming	Python, JavaScript, TypeScript
Scientific Writing	L ^A T _E X, Markdown
Backend / APIs	FastAPI, Django, Express.js
Frontend	React.js
AI / ML	Machine learning, deep learning workflows, reinforcement learning exposure, prompt engineering, LangChain, RAG
Data	EDA, preprocessing, feature engineering, visualization
Deployment	Docker, model serving, deployment concepts, experiment tracking concepts
Automation	Workflow automation tools such as N8N

Structured Training

- freeCodeCamp tracks: Responsive Web Design; JavaScript Algorithms and Data Structures; Front End Development Libraries; Data Visualization; Relational Database; Back End Development and APIs

Languages

Persian	Native
English	Near-native / professional proficiency